

Statistics and data science education research

Preparing to Teach
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A little about me

- BS in Mathematics, MS in Statistics, PhD in Statistics
- Associate Professor of the Practice in Statistical Science at Duke
 - Teaching (excellence)
 - Service (departmental, university, professional)
 - Scholarship (significant contributions to education, research, or service to profession)
- Research primarily focused on student experience in introductory courses
 - *A validation of the short-form classroom community scale for undergraduate mathematics and statistics students*
 - *A Mixed-Methods Study of Classroom Community Among Undergraduate Students in Introductory Mathematics and Statistics*
 - *Evaluating Student Perceptions of Assessment in Introductory Quantitative Studies*

What is statistics/
data science
education
research?

How is this
research done?

What else do I
need know?

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Why do education research?

- ★ Investigate meaningful and high-impact research questions
- ★ Contribute to innovations in teaching
- ★ Research alongside classroom practice
- ★ Improve your teaching
- ★ Collaborate with others interested in statistics/data science education
- ★ Satisfy criteria for promotion and tenure

Scholarship of Teaching and Learning (SoTL)

“Systematic and informed approach to investigate teaching and learning, with the goal of improving student outcomes and advancing educational practice” (Terry et al., 2026)

Characteristics (Felten, 2013)

- Inquiry focused on student learning or experiences
- Grounded in local and scholarly context
- Study is methodologically sound
- Public dissemination of results

Research

“Generating new knowledge and/or innovation through scientific inquiry” (Terry et al., 2026)

Characteristics (Cross, 1999; Terry et al., 2026)

- Identifies problem or gap in literature
- Seeks to acquire new knowledge
- Grounded in literature to motivate research questions and provide theoretical framework
- Study is methodologically sound
- Public dissemination of results

Example of SoTL: van Es et al. (2025)

Inquiry focused on student learning: “investigates student expectations, preferences, and concerns as they start an introductory statistics course at Cornell University”

1. “assess student mindset when they first sit down in the classroom”
2. “investigate the intersection of their predeterminations with socio-demographic variables”
3. “present resources that could help instructors use these insights to enhance student learning”

Example of SoTL: van Es et al. (2025)

Grounded in context:

- **Local context:** Introductory statistics course at Cornell University (245 students)
- **Scholarly context:** Research on students' attitudes, anxiety, and motivation in introductory statistics

Methodologically sound:

- Data collected through online survey administered after the first day of class
- Descriptive statistics used to analyze survey questions and compare subgroups

Example of SoTL: van Es et al. (2025)

Public dissemination of results:

- Article published in *Journal of Statistics and Data Science Education*
- Visual representation of results
- Table to summarize recommendations

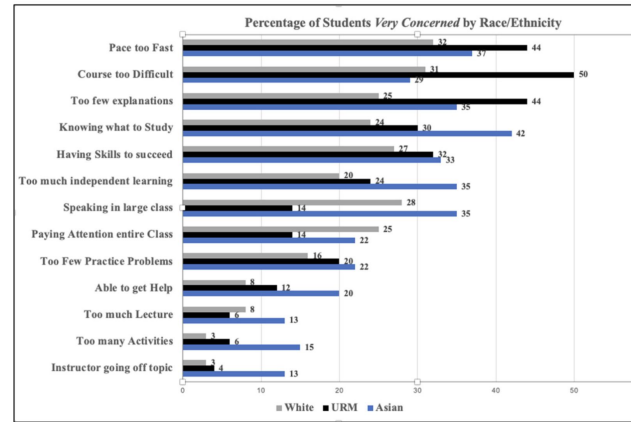


Figure 3. Side-by-side bar graph displaying the percentage of each Racial/Ethnic category who responded Very Concerned for each of the 13 categories.

Pace of the course would be too fast Course would be too difficult Receiving too few in-depth explanations Knowing what to study Having necessary skills/background to succeed	• "How to Teach a Good First Day of Class" Lang (2019). • "Measuring the Promise: A Learning Focused Syllabus Rubric," Palmer, Bach, and Streifer (2014) • <i>McKeachie Teaching Tips</i> (2014) • Learning and motivation in the postsecondary classroom. Svinicki (2004, pp 124–133). • "Structure Matters: Twenty-one Teaching Strategies to Promote Student Engagement and Cultivate Classroom Equity", Tanner (2013)
Focus on specific groups	
Women	• "Impact of Non-Cognitive Interventions on the Performance of Women, Minority and International Students in Introductory Statistics", Van and Shinde (2018)
URM	• "Empirically Validated Strategies to Reduce Stereotype Threat" Walton, Cohen, and Steele (2012) • "Recursive processes in self-affirmation: Intervening to close the minority achievement gap," Cohen et al. (2009). Włodkowski and Ginsberg (1995). A Framework for Culturally Responsive Teaching.
First Generation	• "The Unwritten Rules of Engagement: Social Class Differences in Undergraduates Academic Strategies," Yee (2016) First-Generation College Students as Academic Learners: A Systematic Review. Ives and Castillo-Montoya (2020)
International	• "Enhancing International Students' Experiences: An Imperative Agenda for Universities in the UK," Ramachandran (2011)

van Es, C. L., Meaders, C. L., & Smith, M. K. (2025). Examining Initial Expectations, Preferences, and Concerns of Diverse Students in an Introductory Statistics Course to Inform Course Improvements. *Journal of Statistics and Data Science Education*, 1-13.

Research example: Justice et al. (2020)

Gap in literature: “...but as yet does not explore additional dimensions of conceptions held by students at the introductory level”

New knowledge:

1. “What are aspects of students’ conceptions of statistics and statistical thinking?”
2. “How do students’ conceptions of statistics and statistical thinking reflect experts’ conceptions?”

Grounding in literature:

- **Motivation:** Previous work on experts’ conceptions of statistics (Wild and Pfannkuch, 1999)
- **Theoretical frameworks:** Grounded theory and Phenomenology

Justice, N., Morris, S., Henry, V., & Fry, E. B. (2020). Paint-by-Number or Picasso? A Grounded Theory Phenomenographical Study of Students’ Conceptions of Statistics. *Statistics Education Research Journal*, 19(2), 76-102.

Research example: Justice et al. (2020)

Methodologically sound:

- Survey instrument developed through iterative process with feedback from experts
- Data analyzed through multiple rounds of qualitative coding, two independent round and one round of researchers coming to consensus
- Results include themes regarding students' conceptions of statistics, connections to experts' conceptions, and proposed framework of conceptions

Research example: Justice et al. (2020)

Public dissemination of results:

- Article published in *Statistics Education Research Journal (SERJ)*
- Visual representation of results
- Painting metaphor to illustrate conceptions in each quadrant of the framework

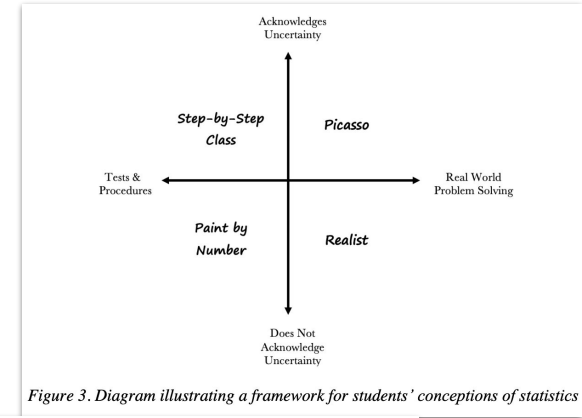


Figure 7. A meaning-making image with uncertainty (the precise appearance of the model is indiscernible). Reprinted under Creative Commons 2.0 license (<https://creativecommons.org/licenses/by/2.0/>) with attribution: P. Picasso, 1935, *Head of a woman*. [Oil on Canvas]. Sotheby's, London.

Justice, N., Morris, S., Henry, V., & Fry, E. B. (2020). Paint-by-Number or Picasso? A Grounded Theory Phenomenographical Study of Students' Conceptions of Statistics. *Statistics Education Research Journal*, 19(2), 76-102.

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Conducting research

- ✓ Identify the research question
- ✓ Review the literature
- ✓ Design the research study
- ✓ Prepare the Institutional Review Board proposal
- ✓ Conduct the study
- ✓ Disseminate results

Identify the research question

- Get inspiration from your own teaching practice
 - *What are you curious about? What challenges are you facing ?*
- Read the statistics/data science education literature
 - *What questions and topics are others researching? What are the open questions?*
- Attend statistics education talks/ conferences/ webinars
 - *What are the current questions being asked? How can you contribute?*

Review the literature

- Know what has already been published and how your study can address a gap in the research or show an alternative perspective
- Identify theoretical framework to ground your work
- Start with journals dedicated to...
 - Statistics education (more on these later)
 - Mathematics education (e.g., [PRIMUS](#))
 - Computer science education (e.g., [SIGSE Conference Proceedings](#))
 - Learning sciences (e.g., [Journal of the Learning Sciences](#))

Current research topics

- Computing in statistics/ data science courses
- Teaching beyond the first course
- Teacher preparation
- Student and teacher attitudes
- Statistics/ data science in K - 12
- Culturally-responsive teaching
- Data literacy
- Teaching diverse learners
- Teaching, learning, and assessment in the age of AI
- ... and others

Design the research study

Experiment

- Best for studying questions about cause-and-effect relationships
- Challenges:
 - Students often self-enroll
 - Ethical considerations of different learning experiences
 - Unavoidable confounding variables
- Quasi-experiments are common

Observational study

- Useful for questions about associations
- Often easier to implement
- Two types of designs
 - Cross-sectional
 - Longitudinal
- Be mindful of confounding variables

Types of research studies

Quantitative

- Useful for answering “what” research questions
- Data typically collected through surveys, evaluating student work, measuring activity (e.g., attendance)
- Consider how to measure variables of interest

Qualitative

- Useful for answering “how” or “why” research questions
- Data typically collected through interviews, focus groups, open-ended survey questions, evaluating-student work
- Ensure research questions can be addressed by data collection method

Qualitative data analysis

- Multiple techniques for analyzing qualitative data (content analysis, thematic analysis, discourse analysis, etc.)
- Demonstrate rigor by addressing three questions (Gordon et al., 2010)
 - *Why would you (or others) trust the findings of this particular research project?*
 - *How does this particular research orientation address the issues of reliability and validity?*
 - *Can the results of this project, involving people in one situation, be transferred to another?*

For examples, see [SERJ Special Issue: Qualitative Approaches in Statistics Education Research](#)

Prepare Institutional Review Board proposal

- Human subjects research that requires review by your institution's Institutional Review Board (IRB)
- IRB proposal typically addresses...
 - Consent
 - Transparency
 - Incentives
 - Potential stress or discomfort to participants
 - Privacy and data storage
 - Technology and AI
- Must include all study materials (recruitment docs, consent form, survey instrument, interview protocol, etc.) in proposal

Disseminate results

Statistics and data science education journals

- *Journal of Statistics and Data Science Education (JSDSE)*
- *Technology Innovations in Statistics Education (TISE)*
- *Teaching Statistics*
- *Statistics Education Research Journal (SERJ)*
- Teacher's Corner of *American Statistician*

Disseminate results

Statistics and data science education conferences

- [Electronic Conference on Teaching Statistics \(eCOTS\)](#)
- [Researchers of Statistics Education \(RoSE\) E-Conference](#)
- [United States Conference on Teaching Statistics \(USCOTS\)](#)
- [Joint Statistical Meeting \(JSM\)](#)
- [International Conference on Teaching Statistics \(iCOTS\)](#)
- [International Collaboration for Research on Statistical Reasoning, Thinking and Literacy \(SRTL\)](#)

Disseminate results

- **Magazines**

- Taking a Chance in the Classroom column in [Chance Magazine](#)
- [Notices of the American Mathematical Society](#)

- **Blogs**

- [Statistics Teaching and Learning Corner](#)
- [Teach Data Science](#)
- [Ask Good Questions](#)
- [Data Pedagogy](#)

- **Podcasts**

- [The Data Science Education Podcast](#)

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Working with collaborators

Benefits of collaborating

- Diverse perspectives, skills, and expertise brought to project
- Share technology and other resources across institutions
- Expand professional network
- Accountability and momentum
- Makes research more enjoyable!

Finding research collaborators

- Reach out to colleagues
- Reach out to people doing work you're interested in
- Meet people at conferences and events
- Connect through online networks such as [Consortium for the Advancement of Undergraduate Statistics Education \(CAUSE\)](#) and [Researchers of Statistics Education \(RoSE\) Network](#)

Working with collaborators

Center for teaching and learning

- Help with study design
- Conduct data collection
- Deidentify data

Students

- Provide valuable perspective to research
- Opportunity to mentor future researchers
- Compensated through pay or course / program credit
- Increased management responsibilities

Funding

- Internal grants at your institution
- Section on Statistics and Data Science Education Member Initiative Grant
- [Spencer Foundation Research Grants on Education](#)
- National Science Foundation
 - [Improving Undergraduate STEM Education: Directorate for STEM Education \(IUSE: EDU\):](#)
 - [Faculty Early Career Development Program \(CAREER\)](#)
 - [EDU: Core Research](#)
 - [IUSE: Innovation in Two-Year College STEM Education \(ITYC\)](#)

Parting thoughts



Follow your curiosity and work on research questions that excite you



Start small



Make time for research (or scholarship) during the term



Read a lot!



Try new things



Celebrate the wins (even the small ones)!

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References + additional resources

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Thank you!



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